

INFO90002 S1 2026

Assignment 2 - SQL

Due: see LMS

Submission: Via the Canvas LMS

Weighting: 10% of your total assessment (Graded out of 200 marks)

Hospital Management System

Community Hospital needs a system to keep track of hospital operations which are depicted in Figure 1. Note, you can download the figure from LMS which would be another option to zoom in and examine the entities and their relationships.

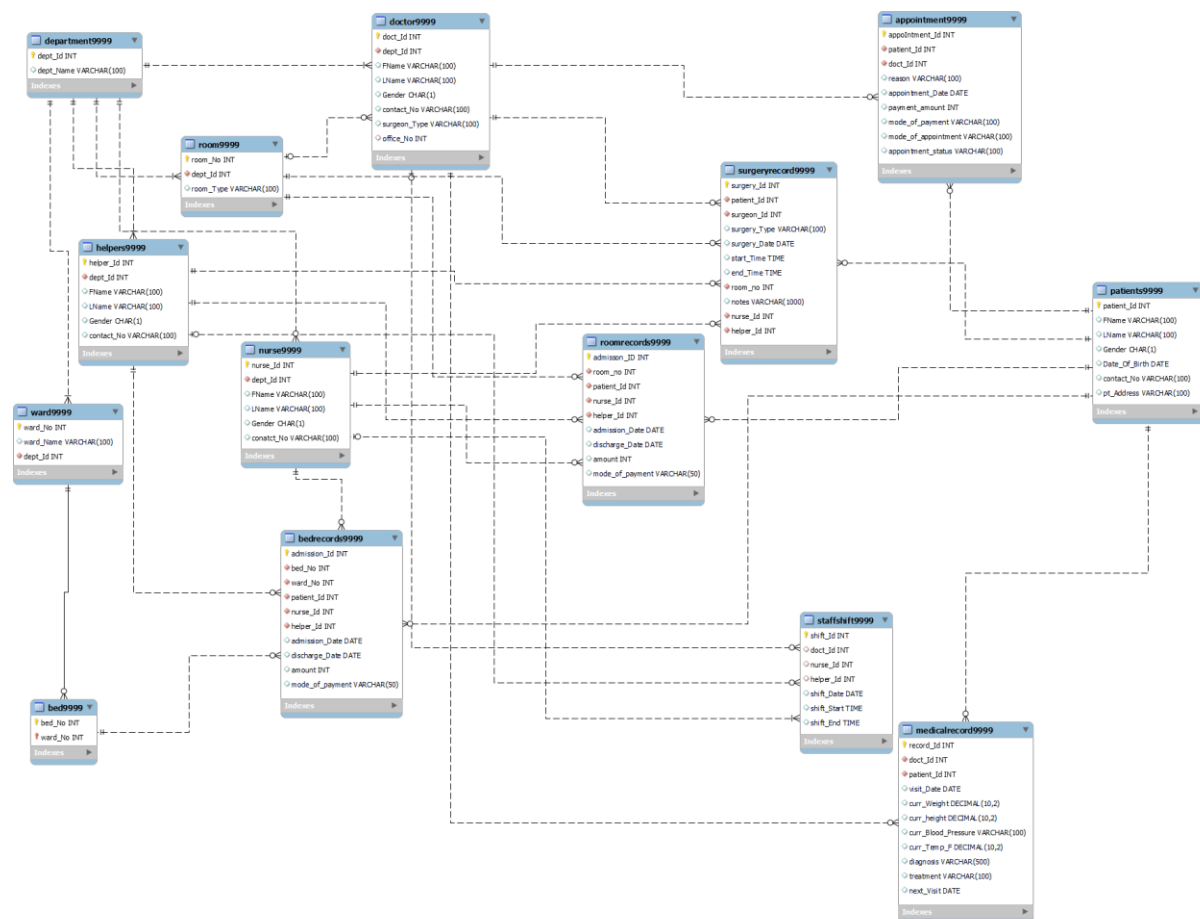


Figure 1. Data Model for Hospital Management System (the diagram is available for download from LMS)

The hospital beds are allocated to wards and wards belong to departments. Each department is in charge of multiple wards.

Each department has multiple doctors, nurses and helpers.

Patients will have their medical records stored in the database.

Patients that are not staying overnight in the hospital (i.e. no beds records for these patients, they are called outpatients) may have appointments with doctors.

Patients that need surgery will have surgery record containing the doctor (surgeon), surgery details, date and start / end time, room where the surgery was performed as well as associated nurse and helper.

Patients who need hospitalisation will have bed allocations.

Instructions

1. **Rename all tables** to have the last 4 digits in table names the same as the last 4 digits of your student ID
 - **Download** the files named **A2Script_2026s1_LOCAL.sql** and **A2Script_2026s1_ENG.sql** from the **LMS**.
 - Open each file in a text editor, e.g. in MySQL Workbench, Notepad, Notepad++ or some other text editor
 - Change all occurrences of **9999** to **the last 4 digits of your student ID** (one way to do this is to perform a **find and replace**). For example, if your student ID is 12349876, your tables will get renamed as room9876, bed9876, ward9876, etc.
Make sure you rename tables in BOTH sql files.
Note, if renaming is not done, you cannot get full marks, a heavy penalty of 50% will be applied.
2. Run the relevant script on the engineering server (and / or on your local MySQL server). This will create all required tables and populate them with data. Note on the local server you will have the database **hospitalmanagementsystem**, while on the engineering server you will only have a collection of tables (possibly mixed with tables from your previous exercises because students do not have permissions for *create database* and *create schema* commands execution).

3. Type the following SQL statement and execute it.

```
SELECT '123459999' as StuID, d.*  
FROM department9999 d
```

(Note that in the above statement 123459999 should be replaced with **your** Student ID and occurrences of 9999 should be replaced with the last 4 digits of your student ID, as in step 1 above)

Notice how each row contains your student id and all attributes from the department9999 table.

StuID	dept_Id	dept_Name
123459999	101	Cardiology
123459999	102	Neurology
123459999	103	Gastroenterology
123459999	104	Nephrology
123459999	105	Pulmonology
123459999	106	Endocrinology
123459999	107	General Medicine
123459999	108	Pediatrics
123459999	109	Orthopedics
123459999	110	Gynecology
123459999	111	Dentistry
123459999	112	Dermatology
123459999	113	Psychiatry
123459999	114	Psychology
123459999	115	Urology
123459999	116	ENT
123459999	117	Surgery
123459999	118	Radiology

You must include your student ID in all SELECT queries. 10% will be deducted from each task where this requirement is not satisfied.

It is expected that

- your script will produce correct results
- your code meets standards of quality as discussed in lectures
- your code runs on the university engineering server

The Task

For each question, write a single SQL statement. **Do not use** inline views / schema on read, views unless explicitly instructed to do so. Views, inline views and schema on read will **earn 0 marks**. Only question 11b is expected to use a view (not inline view).

If your result set is less than 10 rows, show ALL results. If it is longer, show at least 10 rows. Specify how many results were returned in red font under the screenshot.

Note the default setting in Workbench is 1000 records, even if there are many more in a table. So you may need to increase the limit, especially if you get exactly 1000 records returned.

Questions

1. List all types of rooms each department has. Your list should show department ID and name and room type ordered alphabetically by department name.

(10 marks)

2. List all departments, wards they contain and number of beds in each ward that have at least 10 beds per ward. Your list should display department ID, department name, ward number, ward name and total number of beds per ward in the descending order of number of beds.

(15 marks)

3. List doctors who had appointments in January of the last year (i.e. if current year is 2026, the appointments are in January of 2025). Your results should show doctor's full name as first name and last name separated by space, appointment date and patients last name and ID. The results should be ordered by doctor ID and this doctor's appointment dates. Note, your query should work in the future years, not just in 2026.

(15 marks)

4. List doctors with most expensive appointments (i.e. most expensive payment amount). Your results should show doctor's name as last name, comma, first name, then appointment date, then patient name as first name then last name separated by space and finally payment amount of the appointment.

(20 marks)

5. List all wards that currently have no beds. Your results should show ward number and ward name sorted in alphabetical order of ward names.

(10 marks)

6. List how many rooms have never been used within each department, not including consultation rooms. For each department with unused rooms, show the department's id, department's name and the number of unused rooms in the descending order of rooms count.

(20 marks)

7. What is the average length of a surgical procedure, for surgeries where the patient had already presented with diabetes? Show the time in minutes. Hint: use `TIMESTAMPDIFF`. Note: we are interested in all forms of diabetes, so include medical records that mention "diabetes" anywhere in the diagnosis. Also "already presented with diabetes" means the patient had diabetes diagnosed before the surgery date.

(20 marks)

8. List of surgeries that took place where any two of (doctor, nurse or helper) had the same last name. For each of the relevant surgeries, show the surgeon/doctor id as well as the doctor's, nurse's and helper's full names in the format of first and last names separated by space. The results should be ordered by doctor's name.

(20 marks)

9. List numbers of doctors, nurses and helpers in each department and the total number of medical staff (i.e. total of doctors, nurses and helpers in the system). Your list should show department name, medical staff type (as a literal), and number of staff in the department – as shown in the screenshot below. The last row should show empty string in the department name column, the literal 'Total' in the Med staff column and total value in the last column.

StuID	dept_Name	MedStaff	Number
123459999	Cardiology	Doctors	7
123459999	Neurology	Doctors	5
...			
123459999	General Medicine	Nurses	40
123459999	Pediatrics	Nurses	20
...			
123459999	Geriatrics	Helpers	43
123459999		Total	2000

(35 marks)

10.

- a. Add a column `capacity` to the table `room9999`. Provide the SQL statement you used.
Show all columns with the respective data types using the relevant SQL statement.

(3 marks)

- b. Set capacity of all standard rooms to 4. Check that all standard rooms got updated by running a SELECT statement (remember to add your student ID to the SELECT statement).

(7 marks)

11.

- a. Write the SQL DDL to create a view that counts the number of surgeries performed by each surgeon. Your list should include surgeon id, full name as first and last name separated by space and number of surgeries.

Do not include StuID in this view.

In addition to the code, you need to provide 2 screenshots

– the list of tables and views from the left pane of Workbench showing your created view and

– the results of running SELECT from your View (we recommend explicitly showing the SELECT statement used to create the View).

(10 marks)

- b. Using the View you created in Task 11a, list the surgeons who completed the highest number of surgeries. Your results need to display surgeon id, full name and the corresponding number of surgeries.

You **must** include your student ID in task 11b.

(15 marks)

Submission Details:

Submit a single PDF showing your answers to all questions

Please **make sure that you actually submit** your file on LMS. After uploading the file, click on 'Submit Assignment' to actually submit your file. If you submit late because you failed to press the submit button and only noticed this after the deadline, your submission will be considered late just like any other late submission to maintain fairness for all students.

Specify your student name and ID at the top of your answer document.

Formatting requirements for your submission

For each question, present an answer in the following format:

- Show the question number and question in **black** text.
- Show your answer (the SQL statement) in **blue** text (**DO NOT use a screen shot**)
- Show a screenshot of the result from Workbench. If your result set is less than 10 rows, show ALL results. If it is longer, show at least 10 rows (few rows extra is allowed).
- Show how many rows were actually returned, in **red** text.
- Remember to include **your student ID (the only exception is q.11a)**.
- Show each query on a separate page.
- You must not use in-line views, schema on read, views for questions unless explicitly instructed to do so (Q11b only can use a view).
- We may be testing your code on the engineering server so make sure you install the database and test your answers on the "engineering" MySQL server.

Example:

Q.XX List all aircrafts manufactured by Airbus. Your list should show tail number, model, year of manufacturing and manufacturer ordered by the year.

```
SELECT '123459999' as StuID, doct_Id, FName, LName, dept_Id
FROM doctor9999
WHERE dept_Id = 103 OR dept_Id = 105
ORDER BY dept_Id, LName
```

StuID	doct_Id	FName	LName	dept_Id
123459999	1014	Dr. Asif	Abbas Naqvi	103
123459999	1015	Dr. Saba	Asad Ansari	103
123459999	1013	Dr. Saifullah	Mir	103
123459999	1021	Dr. Shahbaz	Ali	105
123459999	1023	Dr. Shazia	Fakhar	105
123459999	1020	Dr. Waqas	Javed	105
123459999	1022	Dr. Abida	Raza	105

7 rows

IMPORTANT: ATTEMPT EVERY QUESTION!

Ensure your scripts run on the engineering MySQL server

Important aspects we consider during marking

- All tables have correct last 4 digits – otherwise deduction from the final mark is 50%
- Student ID appears in all SELECT queries as the first column except question on creating a view or if there is a question where the student's ID is added as part of the INSERT – deduction 10% per question without ID in col 1
- Scripts not tested on the engineering server – 10% deduction from the overall assignment mark
- If the output on the screenshot does not match the SQL script – the mark will be 0. For example, SQL has more or less columns than the screenshot; or the script is not running but there is an attempt to deceive the marker by providing a screenshot.
- Deductions will be made on missing
 - INNER / OUTER in JOIN
 - Aliases, e.g. if the column name contains a function (this example is a real challenge for readability `ROUND(AVG(DATEDIFF()))`)
 - ORDER BY in queries expecting it
- Deductions will be made
 - In cases with hardcoded values, e.g. hardcoding a year will make the query useless in the future
 - When treating data types incorrectly (even if MySQL is forgiving, we are not). For example, marks will be deducted if YEAR is treated as a string.

Please refer to lectures on the expectations of optimised queries that are not wasting CPU time (e.g. avoid using subqueries if a form of JOIN can be used instead).